Chain rule

1 $y = (2x + 3)^3$

2

a $u(x) = 5x^3 - 4x^2 + 3x - 5,\ f(u) = u^7$

b $u(x) = 2x^2 + 2x + 3,\ f(u) = \sqrt[4]{u}$

c $u(x) = 4x^2 - 3x + 5,\ f(u) = \dfrac{1}{u^3}$

3

a $\dfrac{du}{dx} = 8x^3$

b $y = u^5$

c $\dfrac{dy}{du} = 5u^4$

d $\dfrac{dy}{dx} = 40x^3\left(2x^4 + 6\right)^4$

4

a

i $\dfrac{dy}{du} = 5u^4$

ii $\dfrac{du}{dx} = 1$

iii $\dfrac{dy}{dx} = 5(x + 5)^4$

b

i $\dfrac{dy}{du} = -u^{-2}$

ii $\dfrac{du}{dx} = 4$

iii $\dfrac{dy}{dx} = -4(4x + 3)^{-2}$

c

i $\dfrac{dy}{du} = \dfrac{1}{2\sqrt{u}}$

ii $\dfrac{du}{dx} = 2x$

iii $\dfrac{dy}{dx} = \dfrac{x}{\sqrt{5 + x^2}}$

d

i $\dfrac{dy}{du} = \dfrac{1}{3}u^{-\frac{2}{3}}$

ii $\dfrac{du}{dx} = -15x^{-2}$

iii $\dfrac{dy}{dx} = -5 \times 15^{-\frac{2}{3}}x^{-\frac{4}{3}}$

5

a $\dfrac{dy}{dx} = 50x - 70$

b $\dfrac{dy}{dx} = 10(50x - 7)$

6 $\dfrac{dy}{dx} = \dfrac{-2x}{(x^2 + 5)^2}$

7 $\dfrac{dy}{dx} = \dfrac{4}{\sqrt{8x + 5}}$

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8

a $\dfrac{dy}{dx} = 36\left(4x+3\right)^{8}$

b $\dfrac{dy}{dt} = -4\left(2t^{7}+8t^{3}+3t+5\right)^{-5}\left(14t^{6}+24t^{2}+3\right)$

c $\dfrac{dy}{dx} = 3\left(x^{2}+x^{-3}\right)^{2}\left(2x-3x^{-4}\right)$

d $\dfrac{dy}{dx} = \dfrac{1}{3}\left(x^{2}-5x\right)^{-\frac{2}{3}}\left(2x-5\right)$

e $\dfrac{dy}{dx} = 8\left(3x^{2}-4x+2\right)^{3}\left(3x-2\right)$

f $\dfrac{dy}{dx} = -90\left(3x+4\right)^{9}$

g $\dfrac{dy}{dx} = 4\left(9x+7\right)^{\frac{1}{3}}$

h $\dfrac{dy}{dx} = -5\left(x+6\right)^{-6}$

i $\dfrac{dy}{dx} = -\left(3x^{3}-12x^{2}+5\right)\left(x^{4}-4x^{3}+5x\right)^{-2}$

j $\dfrac{dy}{dx} = -\left(1+x\right)^{-\frac{3}{2}}$

k $\dfrac{dy}{dx} = \dfrac{24}{5}\left(4x+1\right)^{\frac{1}{5}}$

l $\dfrac{dy}{dx} = 8\left(\dfrac{1}{3}x+1\right)^{-7}$

m $\dfrac{dy}{dx} = -\dfrac{5}{6}\left(4-\dfrac{1}{3}x\right)^{-\frac{1}{2}}$

n $\dfrac{dy}{dx} = 2\sqrt{5}\left(1-x\sqrt{5}\right)^{-2}$

o $\dfrac{dy}{dx} = 4\left(x^{-\frac{1}{2}}-x^{-\frac{3}{2}}\right)\left(x^{\frac{1}{2}}+x^{-\frac{1}{2}}\right)^{7}$

p $\dfrac{dy}{dx} = \dfrac{4}{3}\left(9-x\right)^{-\frac{7}{3}}$

q $\dfrac{dy}{dx} = -15\left(1-x^{-2}\right)\left(x+x^{-1}\right)^{4}$

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Gradients, tangents and normals

9 $g'(-1) = -256$

10

a $-\frac{3}{2}$ **b** Increasing **c** $-\frac{\sqrt{21}}{3} < x < 0$

11

a $\frac{dy}{dx} = -(3-2x)^{-\frac{1}{2}}$

b $\frac{dy}{dx} = 0$

c C

d The gradient of the tangent gets close to horizontal but never reaches it. There is no value of x where $\frac{1}{\sqrt{3-2x}} = 0$.

12 $f'(8) = 27$

13 36

14 $f'(2) = 3240$

15 $x = 5$

16 $x = 2,\ x = -6$

17 $x = -1,\ 0,\ 1$

18 $x = -2,\ -1,\ 0$

19 $(4, 1)$

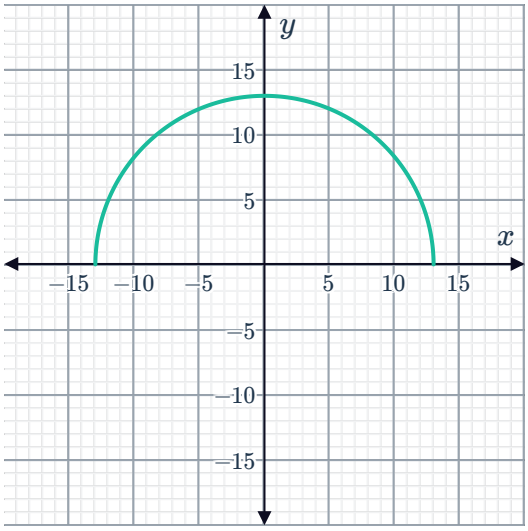
20 $x = 1,\ 6$

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21

a $\frac{dy}{dx} = -x\left(169 - x^2\right)^{-\frac{1}{2}}$

b



c $5x + 12y = 169$

22 $y = -8x - 7$

23 $y = 16x - 15$

24 $y = -\frac{x}{9} - \frac{8}{9}$

25 $y = -\frac{x}{64} + \frac{1025}{64}$

26

a $y = -2x - 1$

b $y = \frac{x}{2} + \frac{3}{2}$

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